Unsupervised Learning

By Matthew Hawkes & Matteo Bloom

Table of Contents

1 Overview

2 Clustering

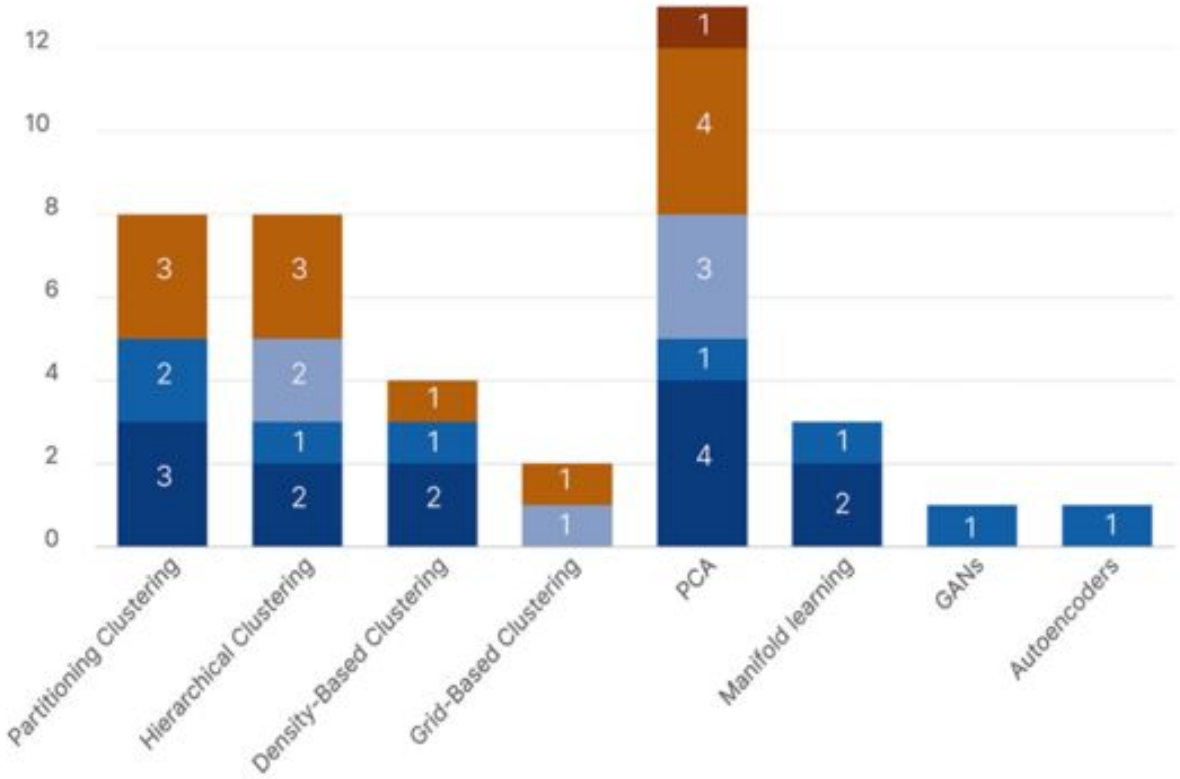
3 Dimensionality Reduction

4 Types of Unsupervised Learning

5 Applications of Unsupervised Learning

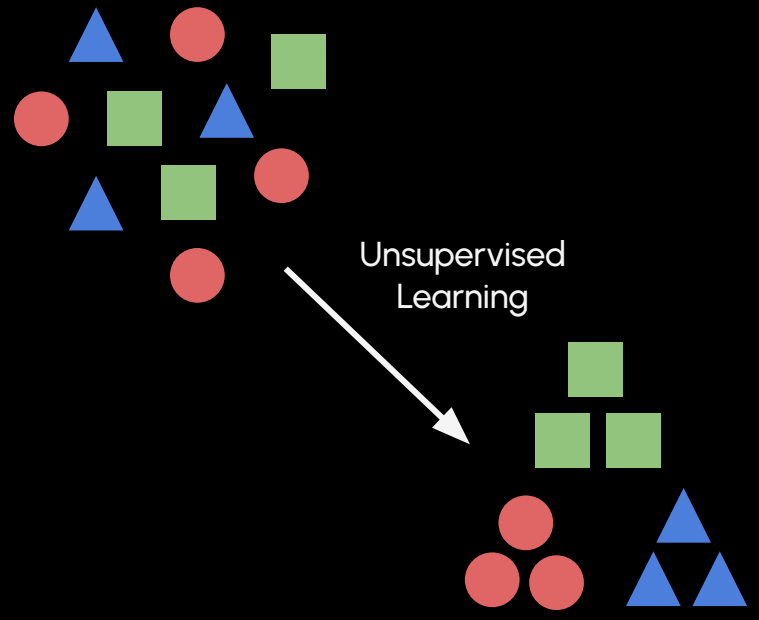
6 Limitations & Conclusion

A bar chart showing all the reported unsupervised machine learning algorithms



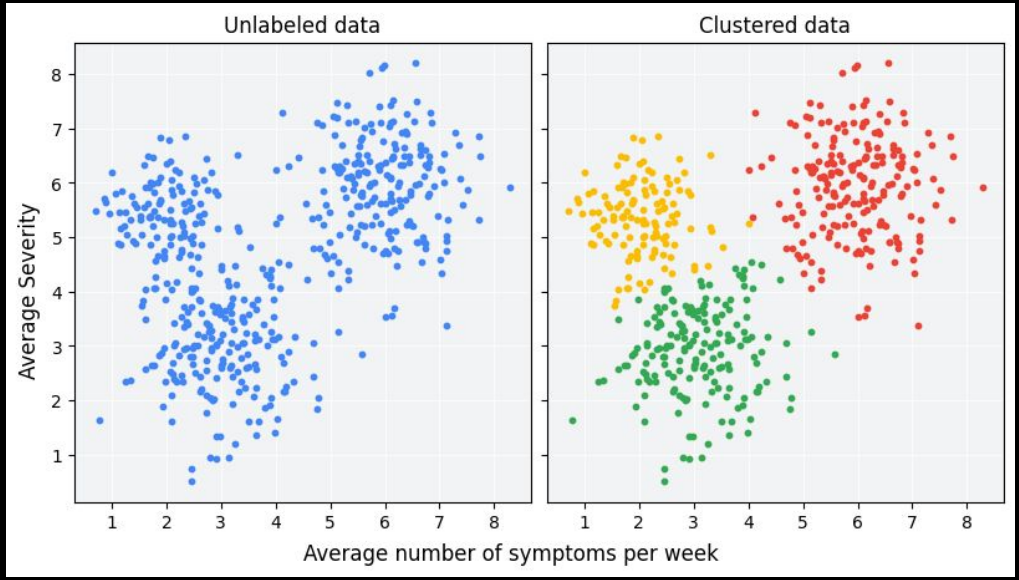
Unsupervised Learning Overview

- A Machine Learning technique that analyzes unlabeled datasets.
- It aims to detect hidden patterns and data groupings.
- It's a great solution to discover similarities and differences in data without knowing labels.



Clustering

- Groups unlabeled data based on their similarities and differences.
- Clustering algorithms are used to process raw, unclassified data into groups.
- Each group is created based on detected structures or patterns in the information



Types of Clustering

- Centroid-Based Clustering:
  - Partitions data into groups based on distance between their centroids
  - Each centroid is either the mean or median of all the points in the cluster
  - K-Means: separates data into 'K' clusters, with each centroid being determined by distance measure.
- Density-Based Clustering:
  - Groups points based on regions of high density.
  - Can discover clusters of arbitrary shape (unlike K-Means).
  - Automatically identifies outliers as noise points.
  - Works well when clusters are irregular or overlapping.
  - ε (epsilon): Neighborhood radius
  - minPts: Minimum number of points required to form a dense region

Types of Clustering

- Hierarchical Clustering:
  - Agglomerative (Bottom-Up)
    - Starts with each point as its own cluster
    - Iteratively merges the closest clusters
    - Produces a dendrogram showing the merge sequence
  - Divisive (Top-Down)
    - Starts with one cluster
    - Recursively splits into smaller clusters
  - Linkage Methods:
    - Single linkage: Minimum distance
    - Complete linkage: Maximum distance
    - Average linkage: Mean distance
    - Ward's method: Minimizes variance
  - Use Cases:
    - Gene Expression Analysis, Customer segmentation, Document Clustering

Dimensionality Reduction

Principal Component Analysis (PCA)
- Linear technique
- Projects data into lower-dimensional space
- Maximizes variance along new axes (principal components)

Why PCA matters:
- Reduces noise
- Speeds up training
- Helps visualize high-dimensional data

Manifold Learning

Non-linear dimensionality reduction methods that assume data lies on a lower-dimensional manifold.

Examples:

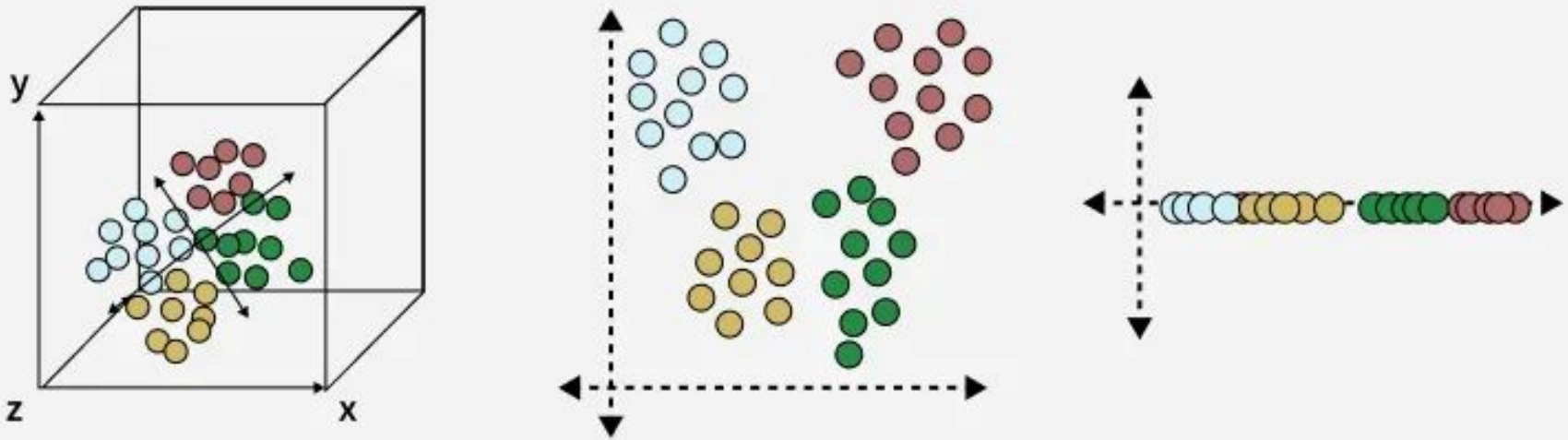
- t-SNE (great for visualization)

- UMAP (preserves global + local structure)

- Isomap

What is Dimensionality Reduction

Dimensionality Reduction is the process of reducing the number of input variables (features) in a dataset while preserving as much important information as possible.



Neural Network-Based Unsupervised Methods

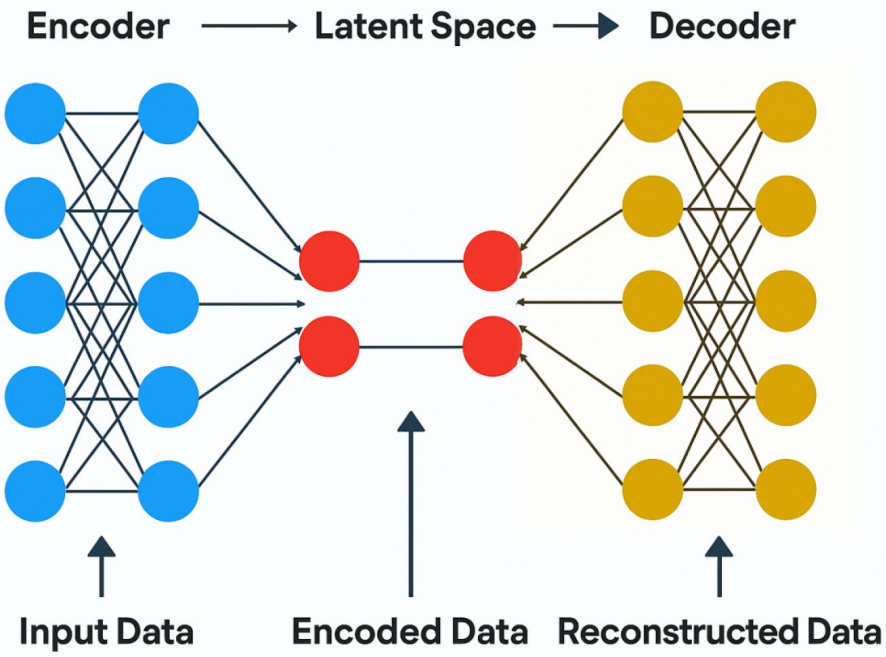
Autoencoders
- Neural networks trained to reconstruct input
- Learn compressed latent representations
- Useful for:
- Anomaly detection
- Denoising
- Dimensionality reduction

Architecture:
- Encoder: Compresses input
- Latent space: Bottleneck
- Decoder: Reconstructs input

Generative Adversarial Networks (GANs)
- Two networks compete:
- Generator: Creates synthetic data
- Discriminator: Distinguishes real vs fake
- Used for:
- Image generation
- Data augmentation
- Style transfer

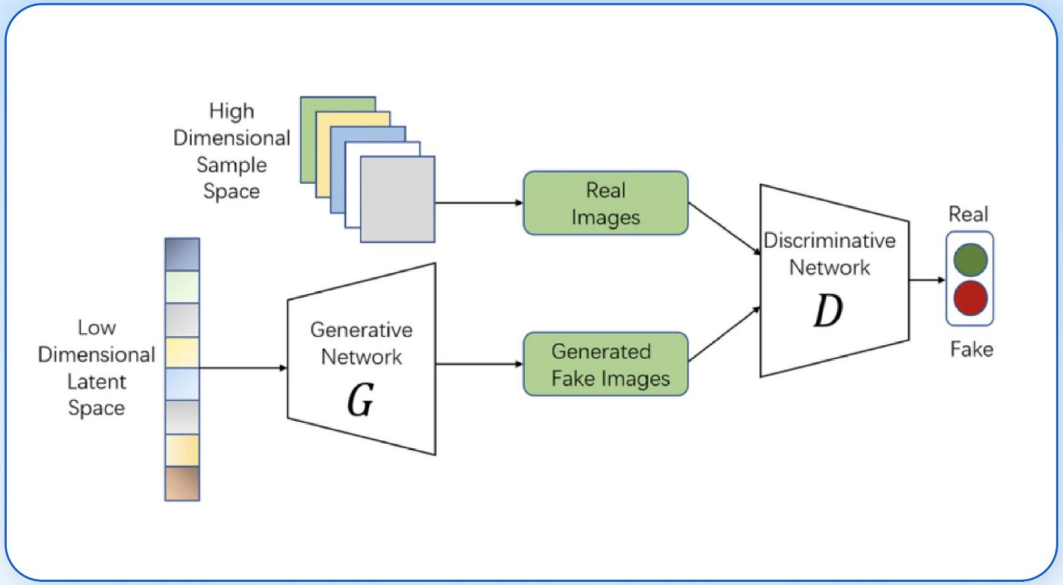
Autoencoders

- The diagram illustrates an autoencoder neural network, showing how input data is compressed into a latent representation and then reconstructed through decoding.
- It highlights the three main components: Encoder (blue nodes), Latent Space (red nodes), and Decoder (yellow nodes), with arrows indicating the data flow from input to output.



Generative Adversarial Network (GAN)

- The diagram shows how a Generative Adversarial Network (GAN) works by having a Generator create fake images from a low-dimensional latent space and a Discriminator evaluate whether images are real or fake.
- It visually contrasts real images from a high-dimensional sample space with generated ones, highlighting the adversarial training between the Generator (G) and Discriminator (D).



XENONSTACK

Real-World Applications of Unsupervised Learning

Cybersecurity

- Detecting anomalous network traffic
- Identifying unknown attack patterns
- Clustering malware families

Healthcare

- Grouping patients by symptoms
- Discovering disease subtypes
- Reducing dimensionality of genomic data

Marketing & Business

- Customer segmentation
- Recommendation systems
- Market basket analysis

Computer Vision

- Feature extraction
- Image compression
- Unsupervised pretraining

Limitations

- No ground truth labels → evaluation is difficult
- Results can be unstable depending on initialization
- Sensitive to scaling and preprocessing
- Hard to interpret clusters in high-dimensional space
- Some algorithms require domain expertise to tune

Conclusion

Unsupervised learning is a powerful set of techniques for discovering hidden structure in unlabeled data. From clustering to dimensionality reduction to generative modeling, these methods form the backbone of exploratory data analysis and modern AI systems. As datasets grow in size and complexity, unsupervised learning becomes increasingly essential for uncovering patterns, reducing noise, and enabling downstream supervised tasks.